Assignment 5

Introduction to Hash Cryptography

Subject: Cryptography and Network Security

Instructions: Answer all five questions. Use suitable examples and explain technical terms clearly.

Q1. Define hash cryptography. Explain the fundamental principles and techniques of cryptographic hash functions with suitable examples.

Answer:
Hash cryptography is a technique that converts input data of any size into a fixed-size value called a hash or message digest. It uses a mathematical algorithm called a cryptographic hash function. The same input always produces the same hash, but even a small change in the input usually produces a very different hash.
Fundamental principles:
1. **Deterministic:** The same message always gives the same hash value.
2. **Fixed-length output:** The output has a fixed size, regardless of input size. For example, SHA-256 produces a 256-bit hash.
3. **One-way operation:** It should be practically impossible to recover the original message from its hash.
4. **Avalanche effect:** A small change in input should cause a large, unpredictable change in the output.
5. **Collision resistance:** It should be extremely difficult to find two different inputs with the same hash.
Techniques: A hash function processes data in blocks using operations such as bitwise operations, shifts, rotations, and modular addition. Padding is added when needed so that the message can be processed in fixed-size blocks. The final result is the message digest. **Example:** A file can be processed using SHA-256 to produce a hash. The receiver calculates the hash of the received file and compares it with the original hash. If both match, this gives strong evidence that the file has not changed.

Q2. Discuss the role of cryptographic hash functions in cryptographic applications. Explain their use in data integrity verification, digital signatures, password storage, and message authentication.

Answer:
Cryptographic hash functions are used in many security applications because they create a compact digest of data. **1. Data integrity verification:** A sender or system calculates a hash of a file or message. The receiver calculates it again and compares the values. If they match, the data is likely unchanged. For example, a downloaded file can be checked using its published SHA-256 hash. **2. Digital signatures:** Instead of signing a large document directly, a system first calculates its hash. The sender signs the digest using a private key. The receiver verifies the signature using the sender's public key and checks the document's hash. This helps detect changes and confirms the signer. **3. Password storage:** Websites should not store plain-text passwords. They store password hashes created using a password-hashing method such as Argon2, bcrypt, or scrypt, with a unique salt. When a user logs in, the entered password is processed and checked against the stored value. Fast general-purpose hashes such as plain SHA-256 alone are not recommended for password storage. **4. Message authentication:** Hash functions are used in constructions such as HMAC. HMAC combines a secret key with a hash function to help verify message integrity and authenticity. For example, two systems that share a secret key can use HMAC-SHA-256 to check that a message has not been modified and came from someone who knows the key.

Q3. Explain the advantages and disadvantages of hash cryptography. Discuss the major properties of a secure cryptographic hash function, including preimage resistance, second-preimage resistance, and collision resistance.

Answer:

Advantages:

1. Hashing is fast for many common algorithms and can process data of different sizes.
2. It produces a fixed-size digest, making data easier to compare and verify.
3. It helps detect accidental or intentional changes to data.
4. It supports digital signatures, HMAC, certificates, and other security systems.
5. A secure hash function is designed to make reversing the hash computationally impractical. **Disadvantages / limitations:**

1. Hashing is one-way, so the original data cannot normally be recovered from the hash.
2. Collisions are mathematically possible because many possible inputs map to a finite number of outputs.
3. Older algorithms such as MD5 and SHA-1 are no longer considered secure against collision attacks.
4. A plain hash does not prove who created a message; authentication requires a mechanism such as HMAC or a digital signature.
5. Fast hashes are unsuitable by themselves for password storage because attackers can test many guesses quickly. **Properties of a secure cryptographic hash function:**

Preimage resistance: Given a hash value, it should be computationally infeasible to find an input that produces it.

Second-preimage resistance: Given one particular input, it should be infeasible to find a different input with the same hash.

Collision resistance: It should be infeasible to find any two different inputs that produce the same hash. These properties help protect stored data, messages, and digital signatures from different types of attacks.

Q4. Describe the Secure Hash Algorithm (SHA) family. Explain the development and key features of SHA-1, SHA-2, and SHA-3, highlighting their major differences.

Answer:

The Secure Hash Algorithm (SHA) is a family of cryptographic hash algorithms developed for information security. Different versions produce message digests of different lengths and use different internal designs. **1. SHA-1:** SHA-1 produces a 160-bit hash value. It was widely used in older security systems, but practical collision attacks have shown that it is not secure for applications requiring collision resistance. It should not be used for new security systems. **2. SHA-2:** SHA-2 is a family that includes SHA-224, SHA-256, SHA-384, and SHA-512, among others. SHA-256 produces a 256-bit digest and SHA-512 produces a 512-bit digest. SHA-2 is widely used and is considered secure when implemented correctly. **3. SHA-3:** SHA-3 is a newer family based on the Keccak design. It includes SHA3-224, SHA3-256, SHA3-384, and SHA3-512. Its internal construction is different from SHA-1 and SHA-2, providing an alternative design. **Major differences:**

• **Output:** SHA-1 produces 160 bits; SHA-2 and SHA-3 offer several output sizes, including 256 and 512 bits.
• **Security:** SHA-1 is considered broken for collision resistance. SHA-2 and SHA-3 are recommended choices for modern applications.
• **Design:** SHA-1 and SHA-2 use a similar general compression-function approach, while SHA-3 uses the Keccak sponge construction.
• **Use:** SHA-2 is very widely deployed; SHA-3 is available as a different design for systems that choose it. **Note:** SHA-1 should be avoided for new digital signatures and integrity-sensitive uses. The appropriate SHA-2 or SHA-3 variant depends on the application and its standards.

Q5. Explain the applications of SHA algorithms in cryptographic protocols and information security. Discuss their role in digital certificates, digital signatures, file integrity checking, and secure communication.

Answer:

SHA algorithms are used in cryptographic protocols and information-security systems to create message digests. These digests help detect changes and support authentication and digital-signature processes. **1. Digital certificates:** A digital certificate contains information about an entity and is digitally signed by a Certificate Authority (CA). Hash algorithms are used as part of the signature process to create a digest of the certificate data.

The signature helps a verifier check that the certificate has not been altered and was issued by the trusted authority. **2. Digital signatures:** The sender calculates a hash of the message and signs the digest with a private key. The receiver verifies the signature with the public key and independently hashes the received message. This supports integrity, authentication of the signer, and non-repudiation when used within a valid system. **3. File integrity checking:** A SHA-256 or SHA-3 hash can be calculated for a file. Later, the hash is calculated again and compared with the saved trusted value. A mismatch indicates that the file has changed. This is commonly used when checking downloaded software and backups. **4. Secure communication:** Secure protocols use hash-based mechanisms to help protect data. For example, TLS uses hash functions in its cryptographic operations, including transcript hashing and HMAC-based constructions in relevant versions and cipher suites. These mechanisms help detect tampering and support secure key establishment and communication. **Conclusion:** SHA algorithms are important building blocks in modern information security. They support integrity checking, digital signatures, certificates, and secure communication. Modern systems generally use suitable SHA-2 or SHA-3 algorithms rather than SHA-1.